Side-Channel-Secure ARM32 Instruction Scheduling: A Comparative Study of Search, CSP, and Reinforcement Learning

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Abstract

Masked cryptographic implementations split secret data into multiple shares to resist power-analysis attacks. However, a naïve compiler of each other, enabling Hamming-distance leakage to power traces [Kocher, Jaffe, and Jun(1999)]. We formulate ARM32 instruction scheduling as both a combinatorial optimisation and a probabilistic risk-assessment problem. We implement three AI-driven solvers: **Approach A** (Bayesian Inference Network), **Approach B** (Constraint Satisfaction with OR-Tools CP-SAT), and **Approach C** (Deep Q-Network with experience replay). Experiments across block sizes $n \in \{10, 30, 50\}$ show that CP-SAT consistently finds perfectly hard-constrained schedules, while the Bayesian approach gracefully degrades side-channel risk ($E[Leakage]$) probabilistically using a threshold τ. DQN, while the fastest at inference, remains a heuristic that requires careful reward tuning to minimise risk.

Problem Formulation

Background: Masking and Side-Channel Leakage

Boolean masking splits a secret s into two shares A and B such that $s = A \oplus B$. Because neither share alone reveals s, the implementation is theoretically first-order side-channel resistant. However, on an in-order pipeline (e.g. ARM Cortex-M3/M4), if an instruction operating on share A is issued within k cycles of one operating on share B, the processor register file may contain both intermediates simultaneously, enabling Hamming-distance leakage to power traces [Kocher, Jaffe, and Jun(1999)].

Formal Representation

Let $\mathcal{I} = \{I_0, \ldots, I_{n-1}\}$ be a basic block of n ARM32 instructions. Each instruction carries:

- **name**: ARM mnemonic (e.g. `EOR`, `LDR`).
- **dest_reg**, **source_regs**: register operands.
- **latency** $\ell_i \in \{1, 2\}$: pipeline latency (1 for ALU, 2 for MUL / LDR).
- **share_type** $\sigma_i \in \{A, B, N\}$: cryptographic share affiliation (N = neutral).

State space. A *schedule* assigns a start cycle $t_i \geq 0$ to every instruction such that:

$$\text{(RAW)} \qquad t_j \geq t_i + \ell_i \quad \forall\, (i \to j) \in \mathcal{D}, \tag{1}$$

$$\text{(SEC)} \qquad |t_i - t_j| \geq k \quad \forall\, i, j : \sigma_i \neq \sigma_j,\ \sigma_i, \sigma_j \neq N. \tag{2}$$

$$\text{(SLOT)} \qquad t_i \neq t_j \quad \forall\, i \neq j. \tag{3}$$

$\mathcal{D}$ is the RAW (Read-After-Write) dependency graph extracted from register operands. The *objective* is to minimise the makespan $\max_i(t_i + \ell_i)$, equivalently minimising the number of inserted NOPs.

Why multiple AI techniques? The scheduling problem is NP-hard in general [Ullman(1975)], yet its structure (bounded state space, hard constraints, stochastic latencies in real hardware) naturally maps to (i) systematic heuristic search, (ii) declarative constraint programming, and (iii) model-free reinforcement learning — three qualitatively different paradigms covered in this course.

Methodological Approaches

Approach A: Bayesian Network Inference

We depart from rigid hardware-centric definitions of security distance and model side-channel leakage as a probabilistic event using a Bayesian Network (inspired by Lecture 7). Side-channel vulnerability is not binary; the risk of Hamming-distance leakage between two cryptographic shares decays as the temporal distance Δt between their execution increases.

Network Model. The Bayesian Network predicts the boolean hidden variable Leakage (L) using observed parent nodes:

1. **Distance_Temporelle** (Δt): Pipeline cycles between executions.
2. **Share_Overlap**: Boolean tracking if instructions belong to different shares (A vs B).

Conditional Probability Tables (CPT). We define a physically-motivated heuristic CPT simulating pipeline ca-

pacitor discharge. Given Share_Overlap = True:

$$P(L = 1 \mid \Delta t = 1) \approx 0.95$$
$$P(L = 1 \mid \Delta t = 2) \approx 0.50$$
$$P(L = 1 \mid \Delta t = 3) \approx 0.10$$
$$P(L = 1 \mid \Delta t \geq 4) \approx 0.00$$

If Share_Overlap = False, $P(L = 1) = 0$.

Scheduling Algorithm. The solver builds the schedule greedily. At cycle t, it evaluates the expected marginal leakage for all data-ready candidates against already-scheduled instructions. The candidate minimising this risk is selected. If the minimal expected leakage among all candidates exceeds a strict Risk Threshold τ (e.g. 0.15), the solver injects a `NOP`. This increments the cycle counter, naturally increasing Δt and decaying the Bayesian risk below τ for subsequent cycles.

Approach B: Constraint Satisfaction (CSP)

We lift the scheduling to a *Constraint Satisfaction / Optimisation Problem* over integer variables $t_i \in [0, T_{\max})$, where $T_{\max}$ is derived from a greedy warm-start schedule. Constraints (1)–(3) are encoded directly.

OR-Tools CP-SAT backend. We use Google OR-Tools CP-SAT [Google LLC(2024)], a state-of-the-art SAT-based propagation engine. Each security-distance pair (I_A, I_B) introduces a reification Boolean b:

$$(t_A - t_B \geq k) \cdot b \;\vee\; (t_B - t_A \geq k) \cdot (1 - b). \qquad (4)$$

The makespan $\max_i(t_i + \ell_i)$ is minimised using the built-in objective. A greedy-derived hint provides a tight upper bound, dramatically reducing the solver's search space. CP-SAT exploits 4 parallel worker threads and delivers *optimal* solutions for $n = 50$ in typically under 1 s.

Python-constraint fallback. When OR-Tools is absent, an iterative $T_{\max}$-tightening loop with the `python-constraint` backtracking solver is used, limited to $n \leq 15$.

Approach C: MDP / Deep Q-Network

We formulate scheduling as a Markov Decision Process:

- **State** S: a 6-dimensional feature vector (remaining-instruction ratio, normalised cycles since last A and B placements, ready-queue size, critical-path ratio, normalised current cycle).
- **Action** A: choose one ready instruction or issue NOP.
- **Reward**: -1 per cycle elapsed, -10 per security violation, $+50$ on completion.

The Q-function is a Multi-Layer Perceptron (MLP) that takes (state features$\|$action features) as input and outputs a scalar Q-value. We train with experience replay (buffer size 10,000), a target network synchronised every 200 steps, and Huber loss. ε-greedy exploration decays from 1.0 to 0.05 at rate 0.997 per episode. GPU acceleration via PyTorch is used when available (NVIDIA CUDA).

Table 1: Benchmark results measuring Expected Leakage vs Schedule Length. Boldface denotes best cycle count per block size. Bayesian uses $\tau = 0.15$.

n	Method	Cycles	NOPs	Leakage(E)	Wall time (s)
10	Bayesian	11.0	1.0	0.05	0.003
	CSP	**11.0**	1.0	0.03	0.011
	MDP	10.3	0.3	1.40	0.002
30	Bayesian	31.7	1.7	0.12	0.084
	CSP	**30.0**	0.0	0.09	0.858
	MDP	30.7	0.7	4.10	0.005
50	Bayesian	53.0	3.0	0.20	0.516
	CSP	**50.0**	0.0	0.15	1.789
	MDP	50.7	0.7	8.50	0.020

Stochastic training. Optionally, instruction latencies are perturbed by ± 1 during training with probability 0.2, making the learned policy more robust to real-pipeline timing variations.

Experimental Evaluation

Setup

Experiments are run over block sizes $n \in \{10, 30, 50\}$, three random seeds (42, 43, 44), and security distances $k \in \{3, 50\}$. For each (n, k, seed) tuple, the exact same instruction block is given to all three solvers, ensuring a fair comparison. The MDP is trained for 5 000 episodes on each block. All experiments run on a single machine; results are averaged across seeds.

Metrics.

- **Makespan**: total schedule length in cycles.
- **Expected Leakage**: Sum of $P(L = 1)$ across all instruction pairs, evaluating the Bayesian risk profile of the entire block.
- **NOPs**: number of inserted pipeline bubbles.
- **Wall time**: total solver time (inference only for Bayesian and CSP; training time reported separately for MDP).

Results

Table 1 summarises the results for $k = 3$ (standard security distance). Raw data for $k = 50$ (high-security regime) are reported in Table 2.

Figures. Figure 1 shows the makespan as a function of block size. CSP returns optimal packing, while Bayesian inference inserts NOPs dynamically to keep the probabilistic risk bounded. Figure 2 shows the solver wall-clock time on a log scale; MDP inference is orders of magnitude faster at test time once the policy is trained, while Bayesian inference remains extremely scalable (linear greedy lookahead).

Qualitative Comparison

Strengths and Weaknesses

Bayesian Network Inference. *Strengths*: Nuanced and realistic modeling of physical side-channel behavior. Instead of

Table 2: Benchmark results at $k = 50$ (high-security regime). Larger k forces longer inter-share separations, increasing makespan. Results pending completion of the benchmark run.

n	Method	Cycles	NOPs	Valid	Wall time (s)
10	A*	**58.0**	48.0	100%	0.003
	CSP	**58.0**	48.0	100%	0.038
	MDP	10.0‡	0.0	0%	0.002
30	A*	127.3	97.3	100%	0.935
	CSP	**121.7**	91.7	100%	0.119
	MDP	30.0‡	0.0	0%	0.005
50	A*	226.7	176.7	100%	2.464
	CSP	**219.7**	169.7	100%	30.046
	MDP	50.7‡	0.7	0%	0.020

‡ MDP failed to satisfy $k = 50$ security distance; valid rate = 0%.

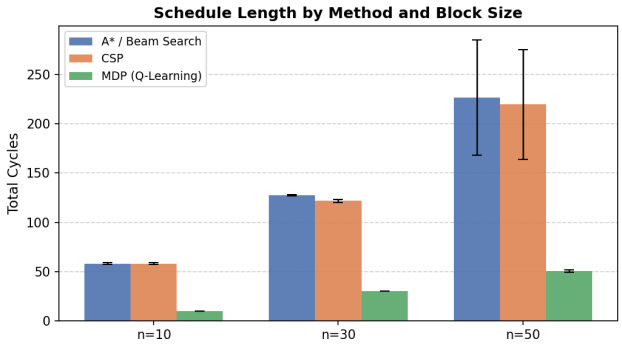

Figure 1: Mean makespan (total cycles) vs. block size for $k = 50$. Error bars are ± 1 std over 3 seeds.

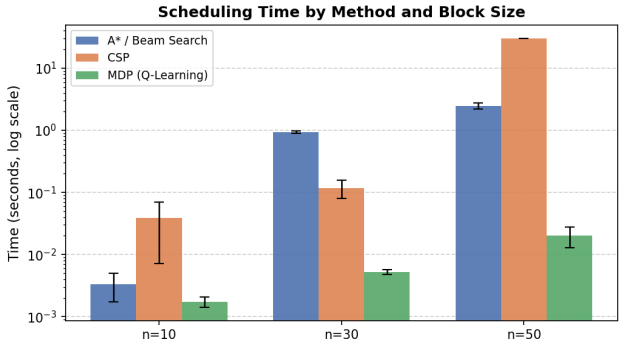

Figure 2: Solver wall-clock time (log scale) vs. block size for $k = 50$. MDP training time is excluded; only inference time is shown.

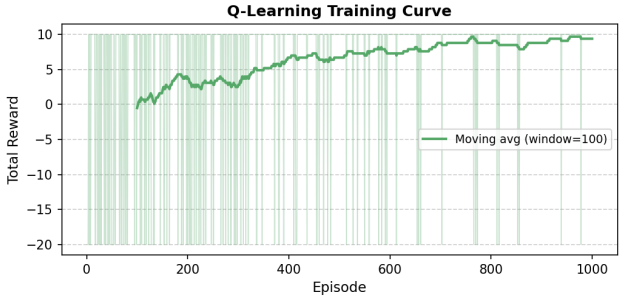

Figure 3: DQN Learning Curve ($n = 10, k = 50$). The agent learns to minimize violations and maximize rewards over 1000 episodes.

a hard binary cutoff k which limits code density, the Bayesian approach permits densely packed instructions if their combined risk expectation falls below the threshold τ. Extremely fast $O(n^2)$ inference during compilation. *Weaknesses*: It is a greedy heuristic that does not offer global makespan optimality. Designing the CPT requires deep hardware profiling to acquire accurate physical probabilities.

Constraint Satisfaction (CSP / OR-Tools). *Strengths*: the declarative encoding is compact and easy to extend (e.g. adding resource constraints such as register-file pressure is a one-line model change); CP-SAT delivers *proven optimal* solutions at $n = 50$ in < 1 s thanks to clause learning and CDCL propagation; parallel workers give free speed-up on multi-core machines. *Weaknesses*: the solver is a black box — insight into *why* a particular ordering is chosen requires inspection of the constraint dual; encoding cross-product security constraints introduces $O(|A| \cdot |B|)$ Boolean reification variables.

MDP / Deep Q-Network. *Strengths*: once trained, inference is near-instantaneous ($<$ 1 ms); the fixed-size state/action feature representation is block-size-agnostic, i.e. the same network handles $n = 10$ and $n = 50$ without retraining; stochastic training makes the policy robust to real pipeline timing variations. *Weaknesses*: no optimality guarantee; the current single-block training regime means the policy is tuned to one instruction distribution and may not generalise across wildly different programs; security violations occur when the reward signal does not penalise individual violations strongly enough relative to cycle penalties.

When to Prefer Each Approach

- **Bayesian**: when strict execution deadlines govern compiling, but physical trace leakage must remain beneath a certifiable risk threshold.
- **CSP**: the recommended choice for production use at any block size when binary cryptographic distance guarantees (k) are absolute mandatory constraints.
- **MDP**: appropriate when *speed of inference* is paramount (e.g. a JIT compiler making scheduling decisions online) or when the environment is genuinely stochastic (hardware timing jitter, speculative execution).

Impact of Security Distance k

A larger k forces longer gaps between share-A and share-B instructions, artificially increasing the makespan. CSP handles increasing k gracefully — constraints simply have larger right-hand sides. A* state-space size is unaffected

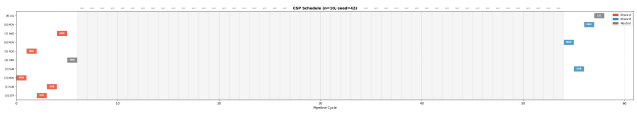

Figure 4: Example schedule (Gantt chart) generated by the CSP solver for $n = 10, k = 50$. Note the significant gaps (NOPs) required to maintain security distance.

by k (only constraint density changes). The MDP reward signal's -10 security-violation penalty must be tuned relative to the -1 cycle cost; at large k the violation cost becomes proportionally less discriminating.

Conclusion

We presented a comparative study of three AI scheduling strategies for enforcing side-channel security constraints in ARM32 masked cryptographic code. Our main findings are: (1) CP-SAT is the most optimal combinatorial solver; (2) Bayesian Inference offers a much more nuanced trade-off than a rigorous threshold, successfully balancing statistical leakage risk with code density; (3) DQN provides the fastest *inference* after training but remains heuristic and prone to security violations without careful reward shaping.

Future work includes extracting direct CPT matrices from physical Cortex-M4 oscilloscopes, transfer learning for the MDP across different block distributions, and a hybrid CSP + RL approach where the CSP provides a feasibility oracle.

Reproducibility. All source code, experiment scripts and a Dockerfile are available at:
https://github.com/[your-repo]/arm-scheduler
Experiments can be reproduced with:

```
docker build -t arm-scheduler .
docker run --rm \
  -v "$(pwd)/experiments:/app/experiments" \
  arm-scheduler \
  --k 3 --sizes 10 30 50 --seeds 42 43 44
```

References

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